

OpenHouse-AI MVP — 5 Week Build Plan

Week	Day	Goal
Week 1 – Data Foundation	Day 1	Project setup (repo, venv, folders, requirements)
	Day 2	Download Zillow + Redfin datasets
	Day 3	Pull mortgage rate data (FRED)
	Day 4	Clean and normalize datasets
	Day 5	Merge datasets into training table
	Day 6	Run exploratory analysis (EDA)
	Day 7	Document dataset + commit pipeline
Week 2 – Valuation Model	Day 1	Define features and target variable
	Day 2	Create training dataset
	Day 3	Train first XGBoost price model
	Day 4	Evaluate predictions
	Day 5	Feature engineering (lags, seasonality)
	Day 6	Retrain improved model
	Day 7	Save model + write model docs
Week 3 – Liquidity Model	Day 1	Define liquidity target (days on market)
	Day 2	Build feature set
	Day 3	Train XGBoost liquidity model
	Day 4	Evaluate error and predictions
	Day 5	Add engineered liquidity features
	Day 6	Retrain + finalize model
	Day 7	Export model and test predictions
Week 4 – Offer Engine	Day 1	Define offer price formula
	Day 2	Implement transaction cost model
	Day 3	Implement holding cost model
	Day 4	Combine valuation + liquidity models
	Day 5	Compute profit estimate
	Day 6	Add risk buffer logic
	Day 7	Test engine on sample homes
Week 5 – Demo Interface	Day 1	Create Streamlit UI
	Day 2	Add property input fields
	Day 3	Connect models to UI
	Day 4	Display offer + profit outputs
	Day 5	Add charts / explanation
	Day 6	Polish UX
	Day 7	Finalize README + demo video